

WEIFANG, China

WMO: 548430

Lat: 36.767N

Lon: 119.183E

Elev: 22

StdP: 101.06

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -10.4	(c) -8.7	(d) -20.5	(e) 0.6	(f) -3.0	(g) -18.3	(h) 0.7	(i) -3.9	(j) 9.4	(k) -1.6	(l) 8.4	(m) -0.9	(n) 2.5	(o) 290	(p) 0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.2	(c) 34.6	(d) 24.2	(e) 33.3	(f) 24.0	(g) 32.0	(h) 23.7	(i) 27.5	(j) 31.8	(k) 26.7	(l) 30.5	(m) 26.0	(n) 29.4	(o) 3.4	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.3	21.9	30.1	25.7	21.0	29.3	25.0	20.1	28.4	88.0	32.2	84.4	30.5	81.0	29.5	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.4	(b) 7.3	(c) 6.3	(e) -12.8	(f) 37.4	(g) 2.5	(h) 1.6	(i) -14.6	(j) 38.5	(k) -16.1	(l) 39.4	(m) -17.4	(n) 40.3	(o) -19.2	(p) 41.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	-1.7	1.2	7.2	13.7	19.6	24.3	27.0	25.9	21.4	15.3	7.3	0.7		
	DBStd	10.51	3.28	3.91	4.59	4.10	3.51	2.69	2.33	2.43	2.89	3.54	4.13	3.41		
	HDD10.0	1129	362	247	113	12	0	0	0	0	4	103	289			
	HDD18.3	2611	620	480	347	146	26	1	0	0	7	105	332	548		
	CDD10.0	2429	0	0	25	124	299	429	527	493	343	168	21	0		
	CDD18.3	869	0	0	1	8	66	180	269	234	100	11	0	0		
	CDH23.3	8338	0	0	12	119	707	1712	2800	2198	708	81	1	0		
	CDH26.7	3159	0	0	2	24	244	737	1188	778	178	8	0	0		
(14) Wind	WSAvg	2.8	2.7	2.9	3.3	3.6	3.2	3.1	2.6	2.2	2.2	2.4	2.6	2.6		
(15) Precipitation	PrecAvg	594	6	13	14	27	50	79	140	142	58	29	26	11		
	PrecMax	920	28	32	40	65	122	180	284	302	162	100	101	36		
	PrecMin	334	0	0	0	3	0	5	14	15	5	2	0	0		
	PrecStd	136	7	10	11	15	29	47	77	77	44	25	26	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	15.7	25.0	29.5	35.0	36.5	36.5	34.5	32.0	27.7	21.5	13.0		
		MCWB	4.2	7.9	13.4	17.6	19.6	20.4	25.6	26.5	22.4	17.2	13.1	7.4		
	2%	DB	7.8	12.6	21.3	26.5	31.4	34.2	34.9	33.0	30.1	25.6	18.7	10.3		
		MCWB	2.7	5.9	11.4	15.9	18.8	21.2	25.6	26.1	21.1	16.6	11.7	5.6		
	5%	DB	5.8	10.4	18.1	24.2	29.3	32.7	33.6	31.8	28.7	23.6	16.2	8.2		
		MCWB	1.4	4.7	9.7	14.8	18.0	21.1	25.7	25.2	20.6	15.7	10.6	4.1		
	10%	DB	3.9	8.3	15.2	21.9	27.3	30.9	32.1	30.6	27.3	21.7	14.2	6.2		
		MCWB	0.3	3.3	7.9	13.7	17.5	20.8	25.2	24.8	20.0	15.1	9.8	2.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.5	9.1	14.7	19.8	22.1	25.7	28.6	28.2	25.8	20.1	14.8	9.1		
		MCDB	8.4	13.4	22.7	26.4	29.5	31.0	33.4	32.5	29.1	23.6	17.8	11.5		
	2%	WB	3.5	7.1	12.4	17.6	20.9	24.2	27.7	27.5	23.7	18.7	13.3	6.6		
		MCDB	6.8	11.6	19.5	24.2	28.1	29.2	32.4	31.4	27.2	22.2	16.8	9.2		
	5%	WB	2.0	5.4	10.5	16.0	19.8	23.2	27.0	26.8	22.4	17.5	11.7	4.7		
		MCDB	5.0	9.2	16.7	22.3	26.5	28.5	31.3	30.2	26.0	21.3	15.0	7.2		
	10%	WB	0.6	3.8	8.8	14.5	18.6	22.3	26.3	26.0	21.5	16.3	10.2	3.2		
		MCDB	3.5	7.7	14.3	20.0	24.9	27.6	30.2	28.9	25.3	20.3	13.9	5.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	10.2	11.6	11.9	11.9	10.8	8.2	7.7	10.0	10.6	10.0	9.1		
		MCDBR	12.6	13.7	15.6	15.6	15.3	14.0	10.6	9.0	11.8	13.3	13.2	12.3		
	5% WB	MCWBR	8.2	8.2	8.1	7.5	6.0	4.6	3.7	3.5	4.3	5.6	7.1	7.9		
		MCWBR	11.0	11.9	14.2	13.8	12.8	10.2	8.4	7.2	9.0	9.9	10.6	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.518	0.577	0.611	0.632	0.654	0.771	0.752	0.690	0.618	0.647	0.584	0.524		
		taud	1.749	1.666	1.622	1.626	1.622	1.467	1.551	1.649	1.733	1.615	1.683	1.765		
	Ebn at Noon		611	632	664	684	679	602	610	636	652	566	545	570		
		Edh at Noon	173	212	242	253	258	301	275	245	213	220	184	161		
(44) All-Sky Solar Radiation	RadAvg		2.37	3.06	4.38	5.13	5.70	5.34	4.72	4.55	4.16	3.32	2.49	2.16		
	RadStd		0.27	0.35	0.32	0.21	0.42	0.39	0.45	0.45	0.34	0.31	0.33	0.20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.74	+1.19	N/A	N/A	N/A	N/A	-97	-161	+179	+99
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+74	N/A

Nomenclature: See separate page